

Guru Saran Kannan

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EDUCATION

University of Massachusetts Amherst

Incoming Fall 2026

M.S. in Computer Science, Amherst, MA

SRM Institute of Science and Technology

Jun 2021 – May 2025

B.Tech in Artificial Intelligence, Kattankulathur, India

CGPA: 7.93/10

TEST SCORES

GRE General (ETS, Apr 2025): 327/340 | V 157 · Q 170 · AW 5.0

TOEFL iBT (ETS, Jul 2025): 106/120 | R 27 · L 30 · S 21 · W 28

SKILLS

Languages	Python, SQL, JavaScript, Lua, Go
Web & Front-end	HTML/CSS/JavaScript, responsive sites (deployed: gurusarank.in), FastAPI, REST APIs, Streamlit, Gradio, Docker; Flask, React, Django (working knowledge)
Databases	SQL/PostgreSQL, DynamoDB, Redis, FAISS (vector DB)
NLP / AI	LLM APIs (OpenAI), LangChain, LangGraph, RAG, HuggingFace Transformers, PyTorch, TensorFlow, Wav2Vec 2.0, text classification, Reinforcement Learning
Data & MLOps	Pandas, NumPy, Scikit-learn, Matplotlib; AWS (ECS/SQS/DynamoDB), Terraform, GitHub Actions CI/CD, Prometheus/Grafana
Tools	Git/GitHub, Vim

PUBLICATION

Virtual Traffic Dynamics (VTD): A Custom AI Traffic Simulation Approach. Proc. Intl. Conf. on Deep Learning and Communications (IconDeepCom 2025), Springer. Presented Apr 2025; forthcoming 2026. [\[CODE\]](#) [\[PAPER\]](#)

PROJECTS

Real-time Recommendation Engine (*In Progress*). PyTorch, FastAPI, Redis, Kafka, PostgreSQL: two-tower neural recommender with a FastAPI serving layer, Redis real-time feature cache, and Kafka event ingestion, targeting sub-20 ms personalized recommendations at ~1K RPS.

LLM Inference Server (async ML serving platform)

Apr 2026

Python, FastAPI, AWS ECS Fargate, SQS, DynamoDB, Terraform, HuggingFace Transformers, Prometheus/Grafana [\[GitHub\]](#)

- Queue-based async inference platform handling 500+ concurrent users at sub-50 ms API latency, decoupling a thin FastAPI tier from a worker fleet.
- Independent worker autoscaling (1→3→6→10) on SQS queue depth; DynamoDB result store with TTL; 8-module Terraform IaC and GitHub Actions OIDC CI/CD.
- 28 unit tests, 0 failures across a 595 k-request load test.

PHIR: Predictive Hybrid Inference Router

Apr 2026

Go, Ollama, Redis, Prometheus, Docker, OpenAI-compatible API [\[GitHub\]](#)

- Middleware routing LLM requests between local edge models and cloud APIs in <10 ms via heuristic complexity scoring; keeps simple calls local to cut cost.
- Drop-in OpenAI-compatible streaming API with edge→cloud context migration, multi-node load balancing, and automatic failover.

Adaptive RAG System for Student Support (Bluearise Technologies)

Aug 2025

Python, LangChain, OpenAI APIs, FAISS Vector DB, Streamlit [\[TESTIMONIAL\]](#)

- Built an adaptive Retrieval-Augmented Generation system that automated 70% of repeat support issues across 500+ student tickets.
- Cut response time from 24 h to under 5 min and lifted satisfaction scores 25% via real-time semantic matching.
- Reached 85% accuracy on unseen ed-tech queries; shipped a Streamlit dashboard for ticket monitoring and performance tracking.

Personal Portfolio & Technical Blog (gurusarank.in)

2025

HTML, CSS, JavaScript, responsive design [LIVE SITE]

- Designed, built, and deployed a fast, responsive personal website (front-end from scratch) showcasing projects, articles, and a technical blog, demonstrating end-to-end web build and deployment.

Intelligent Traffic Signal Network

Dec 2024–Apr 2025

Python, OpenAI Gym, Reinforcement Learning, Unreal Engine [GITHUB]

- RL signal controller: +30% throughput, −22% average wait; validated over 10,000 multi-agent episodes (convergence <75 iterations). Basis of the Springer publication.

Task Planner (LLM agent)

Mar 2025

Python, LangChain, LangGraph, Streamlit [GITHUB] [LIVE DEMO]

- LLM planner that cut daily scheduling time 35% for 50 beta users; 3-node LangGraph workflow (history analysis + Tavily web search + plan synthesis) returning roadmaps in <2 s.

AI-Driven Pronunciation Training (Adaptive Learning Framework)

Jun–Nov 2024

Python, PyTorch, Wav2Vec 2.0, Speech Recognition APIs, LLMs [GITHUB]

- Speech-coaching system used by 120+ ESL learners; +18 pp pronunciation accuracy over six weeks.
- Real-time phoneme/stress detection at ≤120 ms latency; 21% WER baseline trained on 9,000 hrs LibriSpeech/Common Voice.

Toxic Comment Classification (multi-label NLP)

2023

Python, TensorFlow, Scikit-learn, Gradio [GITHUB]

- Trained a 6-label text classifier on 160,000 online comments, reaching micro-F1 0.92 (93% precision / 91% recall).
- Built a Bidirectional LSTM pipeline with TensorFlow TextVectorization (−40% preprocessing time) and a public Gradio demo.
- Relevance: large-scale moral/abusive-language text classification, directly adjacent to the lab's NLP study of morality in language.*

WORK EXPERIENCE

Wasabi Scripts LLC, Software Engineer

Aug 2023 – Present

Remote [WEBSITE]

- Maintain and enhance FiveM game scripts (Lua + HTML/CSS/JS NUI front-ends) used by 31,000+ servers worldwide; underpinned 52,000+ sales.
- Cut critical bug reports 25% via systematic debugging and performance profiling; sustained 99.8% uptime across diverse server environments.
- Drove community engagement, triaging feedback from 200+ active users to prioritize features, in a fully remote, distributed team.

CERTIFICATES

NVIDIA-Certified Associate: Generative AI Multimodal [VERIFY] · IBM AI Fundamentals [VERIFY] · Google Cloud: Prompt Design in Vertex AI [VERIFY] · Hugging Face: AI Agents · Hugging Face: Fundamentals of MCP

SERVICE & OPEN SOURCE

FiveM Open-Source Community, Contributor & Developer Support Volunteer

Aug 2024 – Present

Remote

- Mentored 100+ developers; shipped two Lua resources adopted by 340+ servers worldwide.